

HealthAI Project – Stepwise Checkpoint Upgrade Plan

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This document breaks your transformation into checkpoints so that each feature can be implemented, tested, and confirmed before moving to the next.

CHECKPOINT 1 — Introduce File-Based Hospital Database

- Create folder: hospital_db/
- Create JSON tables:
 - patients.json
 - visits.json
 - vitals.json
 - labs.json
 - diagnoses.json
 - predictions.json
 - notes.json
- Implement backend service: hospital_db_service.py
- Implement basic CRUD for patients.

Goal: Be able to create a patient and fetch their record.

CHECKPOINT 2 — Add Patient Admission UI (Home Page)

- Add “New Patient Admission Form”.
- On submit → POST /patients
- Display generated patient_id
- Add field: “Enter Existing Patient ID” → load profile

Goal: Patient creation from UI must fully work.

CHECKPOINT 3 — Integrate Patient DB Into All Pages

Modify the following pages to accept patient_id:

- Data Page
- EDA Visualisation
- Model Results
- Disease Prediction
- LOS Prediction
- Patient Cohorts
- AI Tools

Each page:

- Should auto-load patient info if patient_id is given
- Should remove manual entry of clinical values

Goal: Full page-to-patient binding.

CHECKPOINT 4 — Modify All AI Tools to Use Patient Records

AI Tools:

- Patient risk assessment
- Notes summarizer
- Image diagnoser
- Feedback analysis
- Future AI chat bot

Each tool must:

- Accept patient_id
- Fetch patient data

- Produce output
- Save predictions/results to predictions.json or notes.json

Goal: AI tools become patient-aware.

CHECKPOINT 5 — Add Visit, Vitals, Labs, Diagnosis Recording

Extend DB service and routes:

- POST /patients/{id}/visits
- POST /patients/{id}/vitals
- POST /patients/{id}/labs
- POST /patients/{id}/diagnoses

Goal: Build a basic EMR-like structure.

CHECKPOINT 6 — Save Model Predictions Back to DB

Each predictive model:

- Disease prediction
- LOS prediction
- Cohort tagging

Must:

- Read patient data
- Run model
- Save model output into predictions.json

Goal: Predictions become part of patient history.

CHECKPOINT 7 — UI Enhancements & Navigation Improvements

- Add “Patient Lookup” section in sidebar
- Improve layout for clinical workflow
- Auto-populate fields using DB

Goal: UI looks like a real clinical app.

CHECKPOINT 8 — Compliance-Oriented Record Structure (Basic)

- Ensure each record is timestamped
- Add created_by, updated_by fields
- Add minimal auditing structure

Goal: Basic compliance alignment.

CHECKPOINT 9 — End-to-End Test Cycle

Verify:

- Creating a patient works
- All pages load patient data correctly
- AI tools integrate with DB
- Predictions persist
- UI flows smoothly

Goal: Confirm stability before upgrading features.

CHECKPOINT 10 — Add Optional AI Chatbot

- Create ai_chat_bot.py
- Add to ai_tools
- Accept patient_id + message

Goal: Provide interactive doctor-patient assistant.

This structured plan ensures each step is stable before moving forward.